Wiring test 1 — connection list

from	to	note
Pico GP2	ULN2803A IN1 (pin 1)	V1 fill A
Pico GP3	ULN2803A IN2 (pin 2)	V2 vent A
Pico GP4	ULN2803A IN3 (pin 3)	V3 fill B
Pico GP5	ULN2803A IN4 (pin 4)	V4 vent B
ULN2803A OUT1–OUT4 (pin 18–15)	black wire V1–V4	red of V1–V4 → +24 V after e-stop
ULN2803A COM (pin 10)	+24 V after e-stop	flyback diodes
ULN2803A GND (pin 9)	GND (adapter minus)	
+24 V adapter	e-stop NC	→ +24 V after e-stop
+24 V after e-stop	main valve 3V210-08	other side of coil → GND
+24 V adapter (before e-stop)	LM7805C IN (pin 1)	1 µF IN–GND
LM7805C OUT (pin 3)	red wire pressure sensors A and B	100 nF OUT–GND
pressure sensor A signal	10 kΩ → GP27, 15 kΩ GP27→GND	100 nF GP27→GND
pressure sensor B signal	10 kΩ → GP28, 15 kΩ GP28→GND	100 nF GP28→GND
potentiometer ends	Pico 3V3 (pin 36) and GND	
potentiometer wiper	Pico GP26/ADC0	100 nF GP26→GND
all GND	Pico GND, ULN pin 9, 7805 pin 2, adapter minus, senso	one common ground